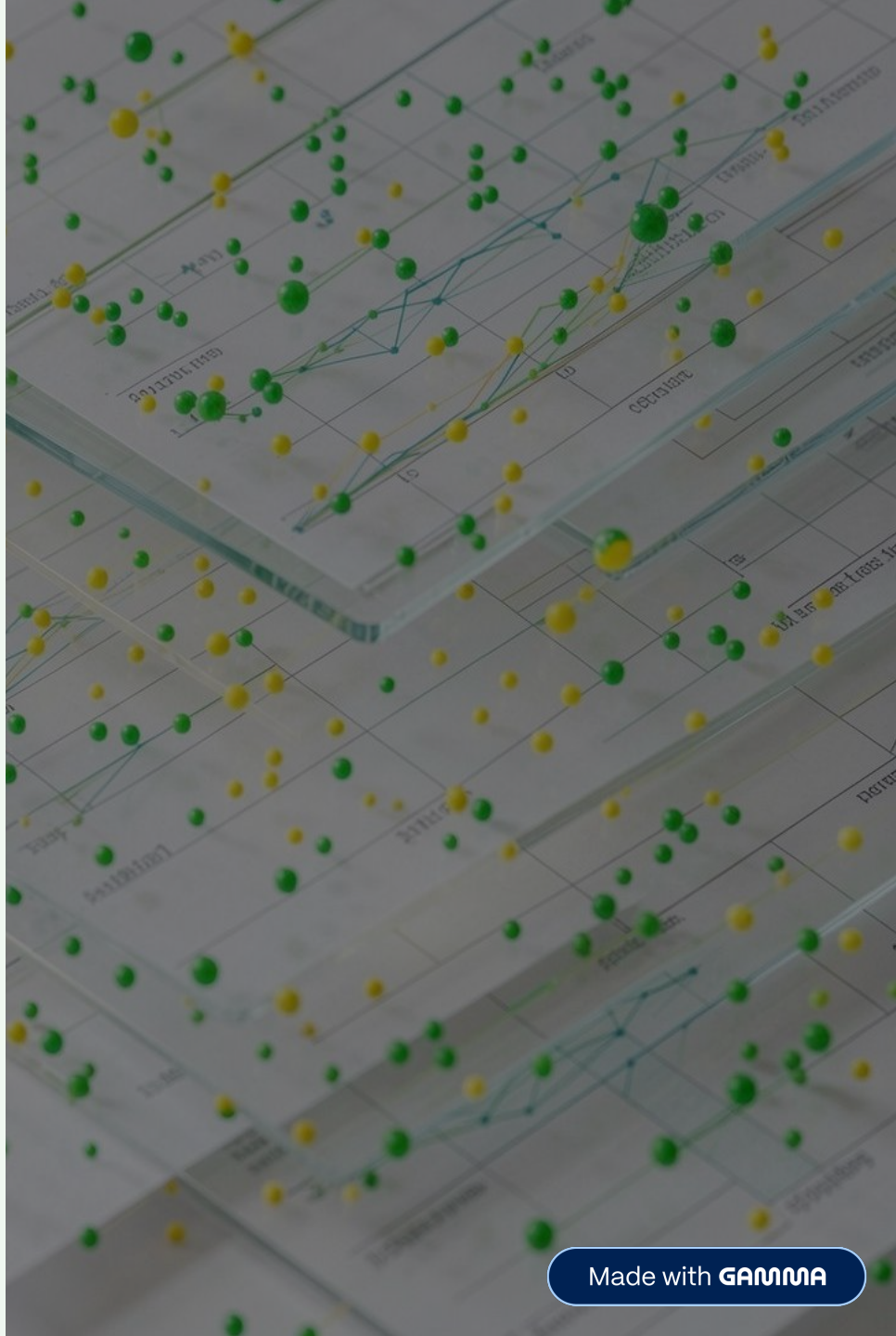
MACHINE LEARNING PROJECT

Multiple Linear Regression for Income Prediction

A focused model using age and experience.



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Project goal: predict income from age and experience

Multiple Linear Regression

Studies the relationship between multiple independent variables and one target variable.

This project

Uses age and experience to predict income.

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02 · DATASET

The dataset is small, focused, and easy to interpret

TOTAL SAMPLES

20

rows

FEATURES

age and experience

TARGET

income



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03 · TOOLCHAIN

Python libraries support the full machine-learning workflow

DATA

numpy + pandas

Data handling and numerical operations.

VISUALS

matplotlib + seaborn

Data visualization.

MODELING

scikit-learn

Train–test split, LinearRegression, RMSE, and R^2 evaluation.

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04 · PREPROCESSING

Before training, the data was separated into X and y

01

Source data

20 rows of age, experience, and income.

02

X · features

Age and experience, the independent features.

03

y · target

Income, the dependent target.

EVALUATION SPLIT

80:20

The dataset was split for unbiased evaluation.

X teaches the model. **y** gives it the answer to predict.

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05 · TRAIN / TEST

16 samples trained the model; 4 samples tested it

TRAINING SET

16

samples used to teach the model

TESTING SET

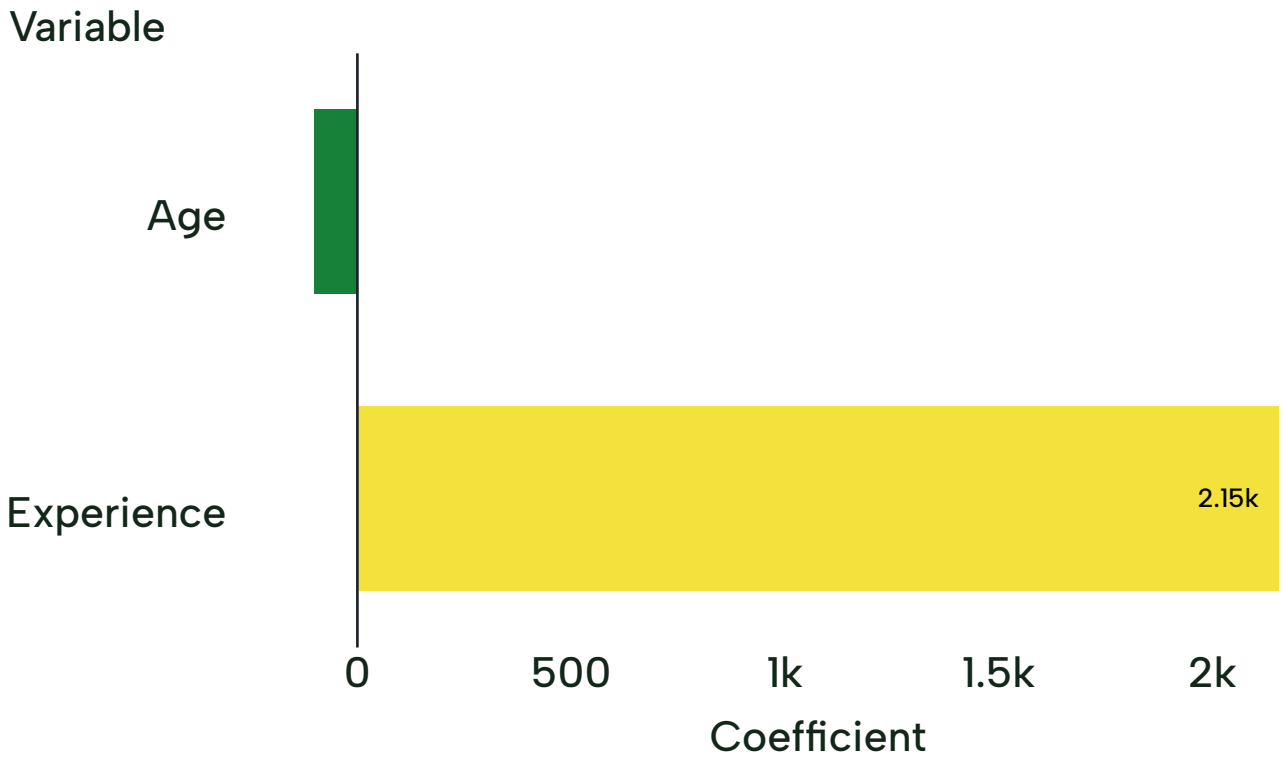
4

samples used to check prediction accuracy

80:20 split · training data builds the model; testing data checks its predictions.

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Experience has the strongest positive coefficient



Age coefficient

−101.0592335

Experience coefficient

2154.80549277

Intercept: 31465.056418503933.

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07 · MODEL LOGIC

The prediction equation summarizes the model logic

$$\text{Income} = 31465.0564 - 101.0592 \times \text{age} + 2154.8055 \times \text{experience}$$

Experience has a strong positive effect.

READ THE TERMS

Intercept
31465.0564

Age
−101.0592

Experience
+2154.8055

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08 · EVALUATION

The model delivered strong performance on test data

RMSE

868.21

Typical prediction error; lower is better.

R^2 SCORE

0.9387

Approximately 93.87% of the data variability explained.

ERROR	EXPLAINED VARIABILITY
868.21	0.9387 · approximately 93.87%

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09 · CONCLUSION

A focused model produced accurate income predictions

Using age and experience, the model achieved strong income-prediction accuracy.

NEXT STEP

More data

Future improvement can come from adding more data.

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